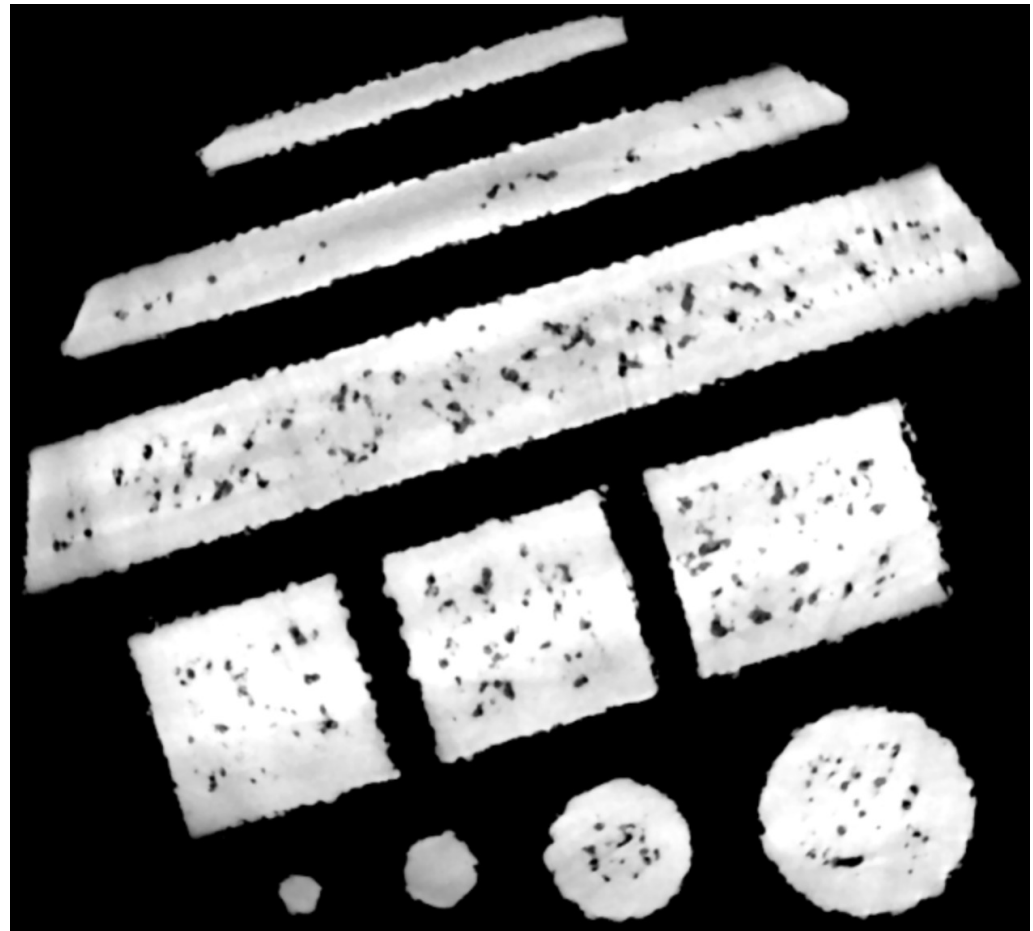
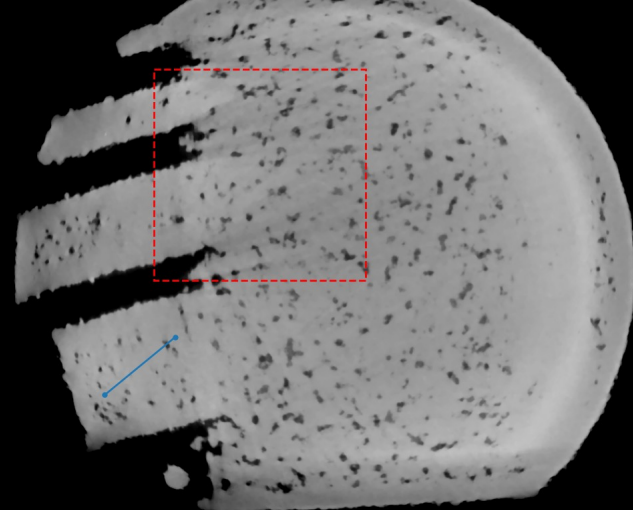


Sample: Inconel parAM sample

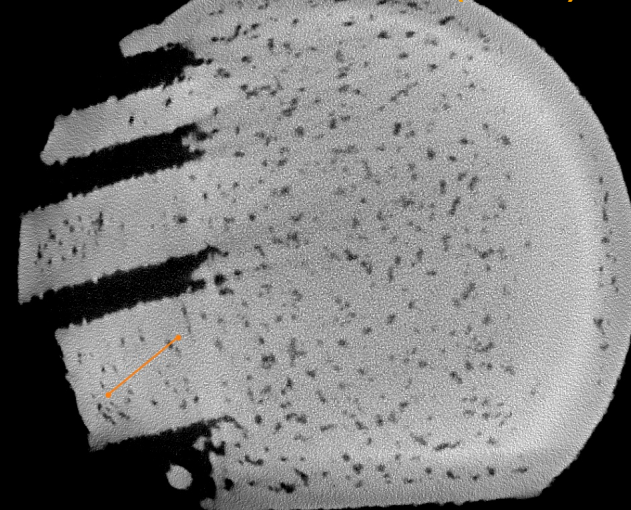
- Scanner: Metrotom
- Recon resolution: 17.3 micron



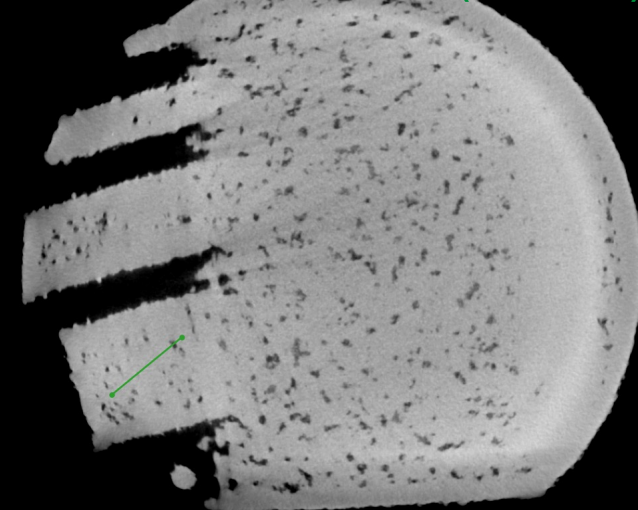
LEAP MBIR 100 iter



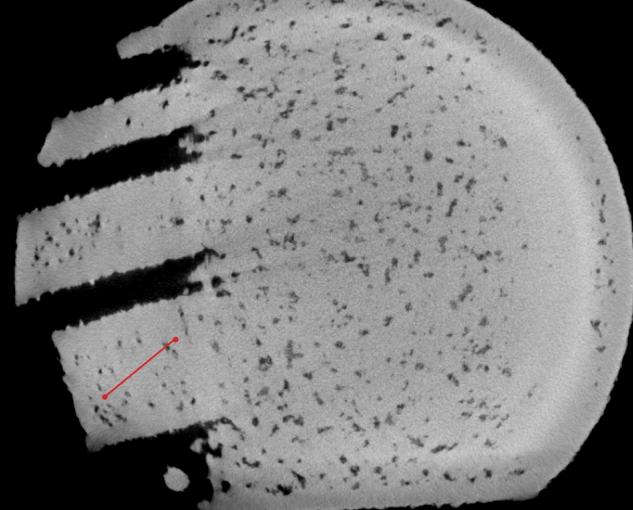
MBIRtorch MBIR 15 iter (init=0)



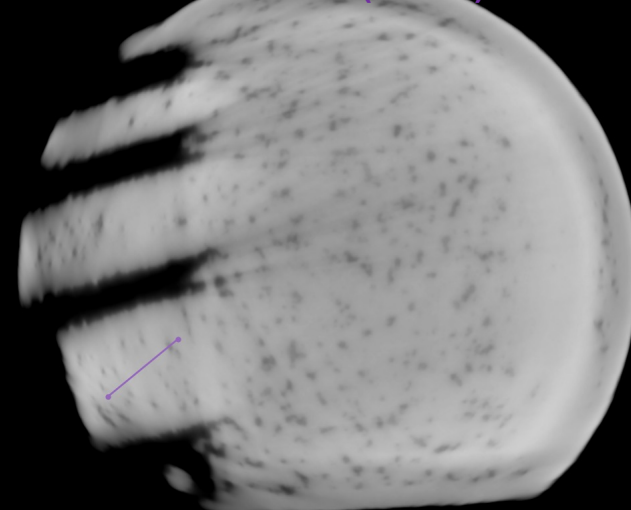
MBIRtorch MBIR 15 iter (init=FDK)



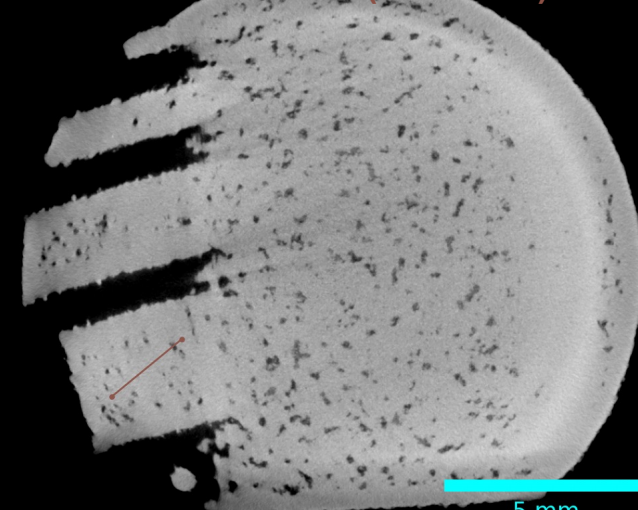
LEAP FDK



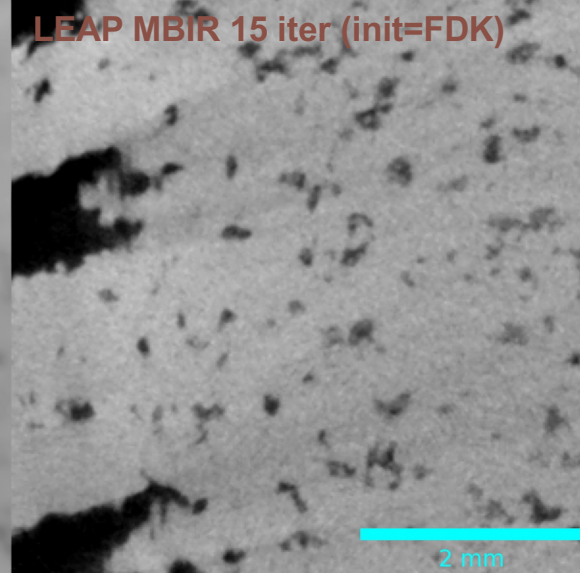
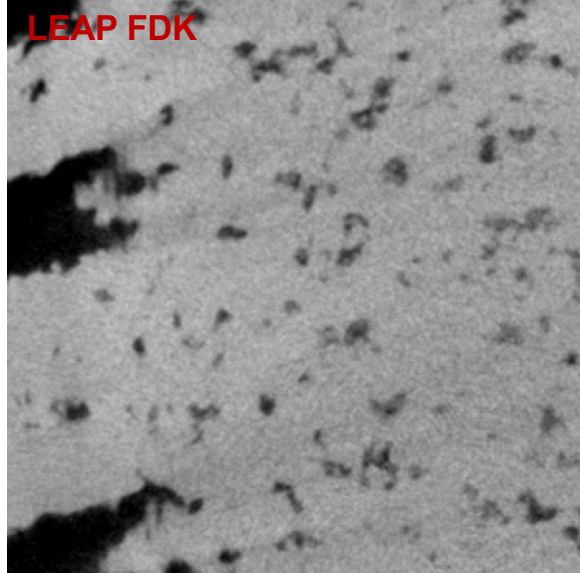
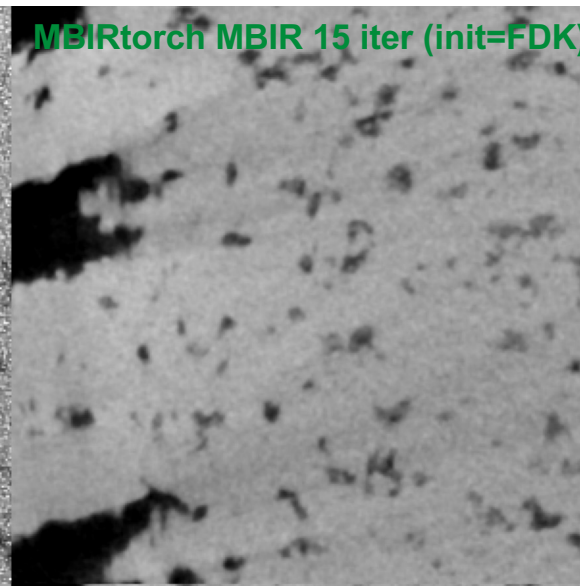
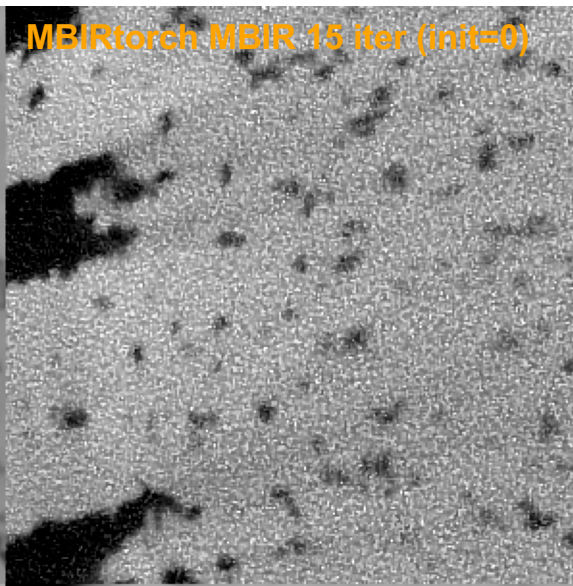
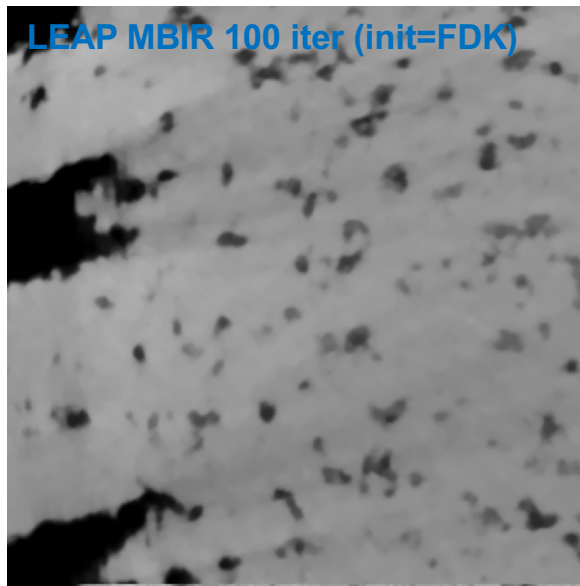
LEAP MBIR 15 iter (init=0)



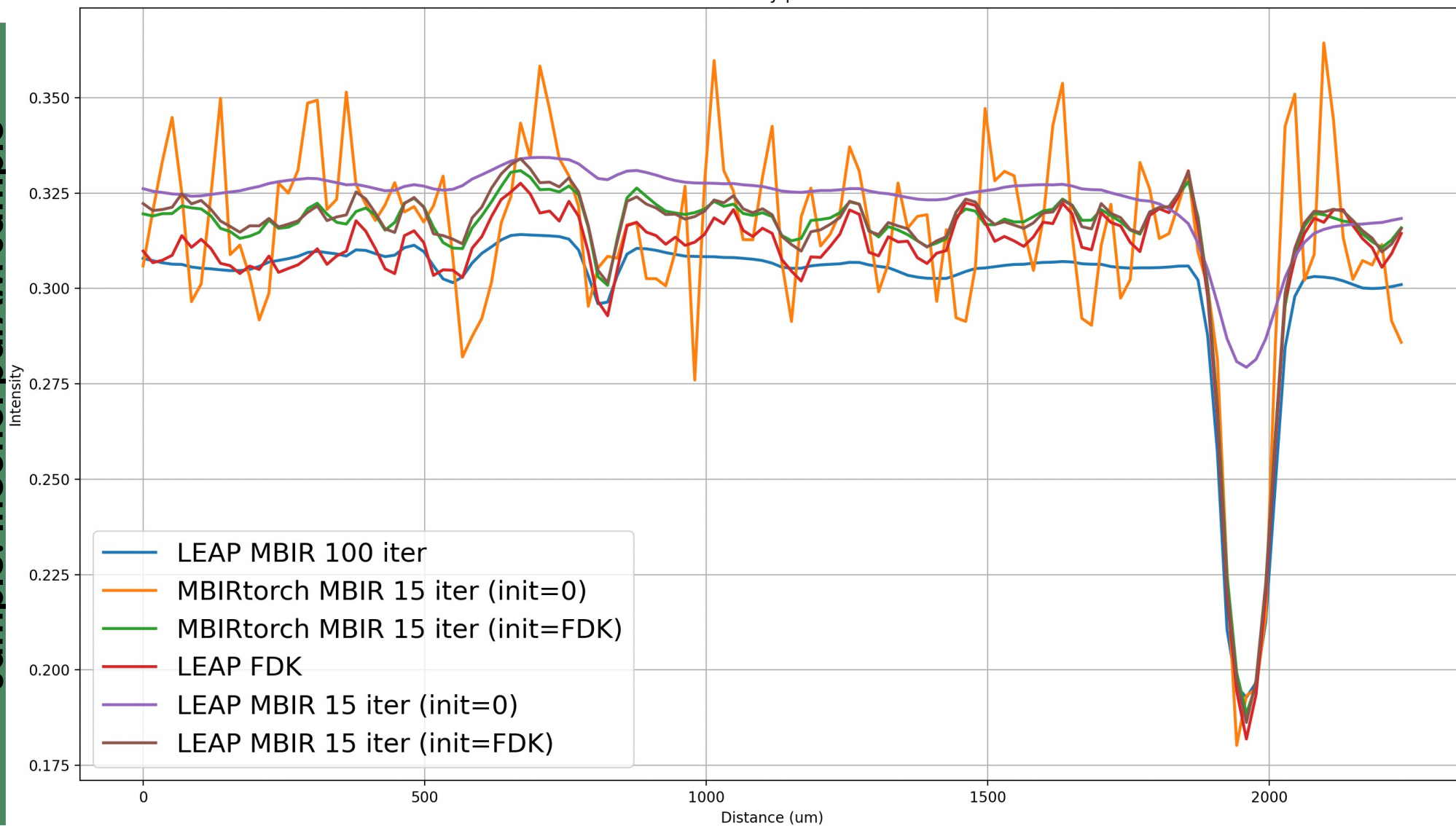
LEAP MBIR 15 iter (init=FDK)



5 mm



Intensity profile



Server: mdf-a100 (A100, 8 GPUs, 40 GB/GPU)

Projection shape (v, r, c): **(2132, 1456, 1840)**

Recon shape (Z, Y, X): **(1296, 1360, 1360)**

GPU count: **4 (A100, 40 GB/GPU)**

CPU memory: **1 TB**

Recon mask: **Disabled**

Recon technique	MBIRtorch*	LEAP
FDK (averaged over 25 runs)	27.31 s	26.78 s
MBIR (15 iterations)	1496.34 s	1666.16 s
MBIR Peak GPU memory (combined)	143.97 GB	23.96 GB
MBIR Peak CPU memory	85.78 GB	175.27 GB

Screenshots

[illegible]

```
Using 4 CUDA device(s) for this reconstruction (was 1).  configure_devices(num_devices=n) pins it.
MBIRtorch FDK 25 times took an average of 27.315849084854126 sec
(mbirtorch) o2x@mdf-a100:~/Projects/LEAP_recon/code$
```

```
FDK recon shape: (1360, 1360, 1296)
MBIRtorch MBIR took: 1496.3457305431366 sec
(mbirtorch) o2x@mdf-a100:~/Projects/LEAP recon/code$
```

```
MBIRtorch MBIR took: 1492.7044911384583 sec
=== GPU Memory ===
GPU 0 peak: 0.51 GB
GPU 1 peak: 0.51 GB
GPU 2 peak: 0.51 GB
GPU 3 peak: 0.51 GB
GPU 4 peak: 35.64 GB
GPU 5 peak: 35.44 GB
GPU 6 peak: 35.44 GB
GPU 7 peak: 35.44 GB
Combined peak (all GPUs): 143.97 GB
=== CPU Memory ===
Peak RSS: 85.78 GB
(mbirtorch) o2x@mdf-a100:~/Projects/LEAP_recon/code$
```

```

det_pix_x, det_pix_y: 0.127 0.127
distance source - rotation axis: 110.080 mm

distance source - detector: 808.814 mm

voxel sizes:0.017285
number of projections: 2132

size: 1186x1261x1199 voxels

recon volume final size:1360 x 1360 x 1296
source location on detector: (926.93, 729.425)

Projection data was read
index      angle      weight      src_shift_x      src_shift_y      src_shift_z      obj_shift_x      obj_shift_y      obj_shift_z

Angle[0] = -0.228452 deg, Angle[-1] = -360.059652 deg, total angles = 2132
(12132, 1456, 1840)
(1456, 1840)
(1456, 1840)
count_data.dtype: float32
brightness.dtype: float64
darks.dtype: float64
subtracting dark from counts
weight_data max, min 31941.8 650.9
normalizing the data with open beam
applying projection offset
norm counts max, min 0.8834313 0.01791063
weight shape = (1456, 2132, 1840)
projection shape = (1456, 2132, 1840)
num_angles = 2132
Regularization parameter = 1673.566757
proj_data shape: (2132, 1456, 1840)
weight_data shape: (2132, 1456, 1840)
LEAP FDK 25 times took an average of 26.78040867805481 sec
MBIR in progress...
Iter 0 / 15 (0.00 s elapsed)
Iter 1 / 15 (111.67 s elapsed)
Iter 2 / 15 (211.44 s elapsed)
Iter 3 / 15 (314.02 s elapsed)
Iter 4 / 15 (414.11 s elapsed)
Iter 5 / 15 (516.77 s elapsed)
Iter 6 / 15 (617.16 s elapsed)
Iter 7 / 15 (720.92 s elapsed)
Iter 8 / 15 (820.98 s elapsed)
Iter 9 / 15 (922.07 s elapsed)
Iter 10 / 15 (1023.29 s elapsed)
Iter 11 / 15 (1127.23 s elapsed)
Iter 12 / 15 (1227.64 s elapsed)
Iter 13 / 15 (1329.90 s elapsed)
Iter 14 / 15 (1429.98 s elapsed)
Done, 15 iterations in 1532.06 s
LEAP MBIR took: 1666.1686294078827 sec
(Leap) o2x8mdf-a100: /Projects/LEAP_recon/condc $

```

```
Done. 15 iterations in 1650.31 s
LEAP MBIR took: 1734.17001080513 sec
=== GPU Memory ===
GPU 0 peak: 0.51 GB
GPU 1 peak: 0.51 GB
GPU 2 peak: 0.51 GB
GPU 3 peak: 0.51 GB
GPU 4 peak: 18.79 GB
GPU 5 peak: 6.16 GB
GPU 6 peak: 6.16 GB
GPU 7 peak: 6.16 GB
Combined peak (all GPUs): 23.96 GB
=== CPU Memory ===
Peak RSS: 175.27 GB
```